Housing Allocation, Fertility, and Population

An illustrative model

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This two-period model illustrates how borrowing limits can keep housing with old owners even when young households would pay more to use it. We show when a planner can improve the allocation by arranging a sale and its financing, holding fertility fixed. We then give a separate condition for fertility to rise and illustrate how temporary fertility changes can leave a permanently larger population.

1 Environment

Households. At date t, the economy contains Y_t young households and O_t old households. A young household has income $y_i > 0$, liquid wealth $b_i > 0$, and chooses total nondurable expenditure c, total housing h, completed fertility $n > 0$, saving $a' \geq 0$, and tenure $m \in \{R, O\}$. Fertility is a continuous choice made once. The pair (y_i, b_i) is drawn from a fixed distribution F in each cohort; estates are warm-glow expenditures rather than the source of the next cohort's b_i.

Preferences. Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of housing. Define the parts of the household bundle available to the adults as

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0. \tag{1}$$

Young and old flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \tag{2}$$

$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \tag{3}$$

Both child requirements reduce the bundle left for adults. All log arguments and preference weights are positive. The estate e is valued in goods at the date after the old period. Parents value what they leave, but their children do not inherit it as entrant wealth in this model.

Demography. A child born at t produces $\nu > 0$ young households at $t + 1$, after survival and household formation. Thus

$$Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t, \tag{4}$$

where $\bar{n}_t$ is average fertility among the young. The conversion from model children to literal children is stated in Section 7.

Housing and asset markets. The fixed housing stock is $\bar{H}$. Rental units satisfy $h \leq h_R^{\max}$, owner units satisfy $h \leq h_O^{\max}$, and $h_R^{\max} < h_O^{\max}$. The consumption good is the numeraire and trades through a one-period bond with price $q \in (0, 1)$ and gross return $R_f = q^{-1}$; only housing clears domestically. The owner of record pays the property tax. A competitive rental intermediary buys a unit for $P_t > 0$ at the beginning of date t, then collects r_t, pays $\tau^p P_t$, and holds a unit worth P_{t+1} at the end of the period. Free entry therefore imposes

$$r_t + P_{t+1} = R_f P_t + \tau^p P_t \quad \Longleftrightarrow \quad r_t = (R_f + \tau^p)P_t - P_{t+1}. \tag{5}$$

Rent is paid in date-$(t+1)$ goods, so its date-t value is $qr_t > 0$; at a steady state, $r = (R_f - 1 + \tau^p)P$. There is no capital-gains tax in the core model. Property tax is retained.

Home finance. A young buyer finances the share $\phi_t \in (0,1)$ and posts $(1-\phi_t)P_t h$ at purchase. Closing occurs before current income and rebates are available, and unsecured borrowing is unavailable. Thus only liquid wealth b_i can satisfy the down payment. The owner cannot buy more housing when old or rent its retained housing to another household. At death the estate sells it back into the fixed stock, and the proceeds enter warm-glow estate expenditure.

2 Household problems

Households take prices and transfers as given. Renters carry financial wealth into old age; owners carry the house and its mortgage. The transfer T_t is the date-t value of the equal rebate of property-tax revenue.

Young renters. Conditional on renting, the young household solves

$$\begin{aligned} W_t^R(i) = \max_{c,h,n,a'} \; & u_t^Y(c,h,n) + \beta V_{t+1}^R(R_f a') \\ \text{s.t.} \quad & c + a' + q\, r_t h = y_i + b_i + T_t, \\ & h \le h_R^{\max}, \qquad a' \ge 0. \end{aligned} \tag{6}$$

The renter enters old age with financial wealth $R_f a'$.

Young owners. Conditional on owning, the young household solves

$$\begin{aligned} W_t^O(i) = \max_{c,h,n,a'} \; & u_t^Y(c,h,n) + \beta V_{t+1}^O(a_{t+1},h) \\ \text{s.t.} \quad & c + a' + [(1-\phi_t) + q\tau^p]P_t h = y_i + b_i + T_t, \\ & a_{t+1} = R_f a' - R_f \phi_t P_t h, \qquad (1-\phi_t)P_t h \le b_i, \\ & h \le h_O^{\max}, \qquad a' \ge 0. \end{aligned} \tag{7}$$

The property tax is set aside in the bond. The mortgage is repaid when the household becomes old, leaving financial wealth a_{t+1} and title $H = h$.

Old households. An old household has financial wealth a. An owner also carries housing H from young age. Renters and owners solve, respectively,

$$\begin{aligned} V_t^R(a) = \max_{c^2,h^2,e} \; & u^2(c^2,h^2,e), \\ & c^2 + qe + q\, r_t h^2 = a + T_t, \quad 0 < h^2 \le h_R^{\max}, \end{aligned} \tag{8}$$

$$\begin{aligned} V_t^O(a,H) = \max_{c^2,h^2,e} \; & u^2(c^2,h^2,e), \\ & c^2 + qe + q\, r_t h^2 = a + P_t H + T_t, \quad 0 < h^2 \le H, \quad e \ge P_{t+1}h^2. \end{aligned} \tag{9}$$

The owner may sell part of its house, and retained housing must be included in its estate. The budget includes sale receipts, property tax, and the value of housing eventually sold by the estate.

Tenure choice. Each household draws once an idiosyncratic preference ξ_i for ownership, independently of (y_i, b_i), and observes it before choosing tenure. It owns when $W_t^O(i) + \xi_i \geq W_t^R(i)$. Assume ξ_i is logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$, which gives the owner share

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \tag{10}$$

The taste draw smooths aggregate tenure shares without changing conditional renter or owner policies.

3 Equilibrium

Prices and transfers equate housing demand to the fixed stock and balance the government budget. Let $\bar{h}_t^Y$ and $\bar{h}_t^O$ denote average housing per young and old household, averaging over types and tenure choices. Let G_t be the old households' distribution over financial wealth, housing, and tenure. Housing clearing and the government budget are

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \qquad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}. \tag{11}$$

The second condition rebates all property-tax receipts in current value. Goods and bonds trade externally; their markets do not impose an additional domestic clearing equation.

Definition 1 (Equilibrium)**.** *Given policies, (Y_0, O_0, G_0), and the fixed entrant distribution, an equilibrium consists of price and transfer sequences, household choices, and cohort distributions such that households solve the stated lifetime and old-age problems using those prices; (5) and (11) hold; and cohort sizes and old states follow from (4) and the preceding young cohort's choices. After an unexpected reform, the initial old cohort retains the assets and housing chosen under its earlier expectations.*

4 The conditional household problem

Fix a household, date, and tenure. Write $A = 1 + \gamma + \omega_B$ and $w_{it} = y_i + b_i + T_t + qT_{t+1}$. With no gains tax, the two tenure-specific user costs coincide: $u_t^R = u_t^O = qr_t$. For the following household comparisons, suppress the type, date, and tenure indices on k_{t+1}^m, $\widetilde{w}_{it}^m$, u_t^m, and $\widehat{h}_{it}^m$, and write $\vartheta = \vartheta_t$.

When saving is positive and the same old-age constraints continue to bind, we can eliminate saving from the young household's problem. Using $s = h - \kappa n$, this gives:

$$\begin{aligned} \max_{x,s,n} \quad & (1+k)\log x + \alpha \log s + \vartheta \log n \\ \text{s.t.} \quad & (1+k)x + us + (\chi + \kappa u)n = \widetilde{w}, \quad s + \kappa n \leq \widehat{h}. \end{aligned} \tag{12}$$

The last budget term is the full resource cost of another child: χ units of goods plus κ units of housing valued at user cost u.

With interior old-age choices, $k = \beta A$ and $\widetilde{w} = w_{it}$. When an old renter's housing limit binds, $k = \beta(1 + \omega_B)$ and the fixed future rent must instead be subtracted from resources:

$$\widetilde{w} = w_{it} - q^2 r_{t+1} h_R^{\max}. \tag{13}$$

These expressions apply while tenure and the other binding constraints remain unchanged. An old owner facing a binding retention or estate constraint requires a different reduction.

At a binding current housing limit, the fertility first-order condition is:

$$\frac{\vartheta}{n} = \frac{\chi}{x} + \frac{\alpha\kappa}{s}. \tag{14}$$

An additional child uses both adult consumption and adult space. The young household's willingness to pay for another housing unit is $MV^Y = \alpha x/s$; the housing constraint binds strictly when $MV^Y > u$.

5 Housing reallocation at fixed fertility

We ask whether the planner can improve the allocation by moving housing from an old owner to a young buyer. Fertility and cohort sizes are fixed, so any gain comes from the use of housing by existing households.

Consider an old owner j with consumption c_j^2, retained housing h_j^2, and estate e_j. Suppose both its retention bound and estate-composition bound are slack. Its retention condition equates willingness to pay to user cost:

$$MV^O = \frac{\gamma c_j^2}{h_j^2} = qr_t. \tag{15}$$

Match it with a young buyer i for whom the down-payment limit is the binding housing restriction and $MV^Y > qr_t$. Each value measures willingness to pay for a small increase in housing, in units of consumption. We compare these values, not marginal utility levels across people.

Let $\epsilon > 0$ be the housing transferred and let $C(\epsilon) \geq 0$ be its real cost in date-t goods. Initially assume $C(0) = 0$ and the right derivative $C'(0) = K_{ijt} \geq 0$. An exact compensating transfer to the old seller is:

$$L(\epsilon) = c_j^2 \left[\left(\frac{h_j^2}{h_j^2 - \epsilon} \right)^{\gamma} - 1 \right] - qr_t\epsilon. \tag{16}$$

After the sale, keeping its estate fixed leaves the seller $qr_t\epsilon$ for consumption; adding $L(\epsilon)$ holds its utility exactly constant. The old retention condition implies $L(0) = L'(0) = 0$ and $L(\epsilon) > 0$ for sufficiently small positive ϵ.

The young buyer borrows to pay for the purchase, compensation, and real cost, then repays the loan when old using the resale proceeds and part of its consumption. The seller replaces the estate's missing housing with a bond. Appendix A records each payment and shows that property-tax receipts and all other households' allocations can remain unchanged.

Proposition 1 (Local allocation gain with explicit compensation)**.** *Maintain the stated interior old-owner conditions, the young buyer's positive saving and slack old-age housing and estate constraints, and slackness of its nonfinancial housing bound. Hold fertility, tenure, cohort masses, prices, common rebates, and other households' allocations fixed. If $MV^Y - qr_t > K_{ijt}$, the stated transaction is feasible for all sufficiently small $\epsilon > 0$, leaves the old seller exactly indifferent, and makes the young buyer strictly better off. All other households, estates, and lenders can be left at their baseline allocations or returns.*

The proof is in Appendix A.

The gain requires the planner to extend credit at purchase. The result does not establish an improvement under the original private credit limit or say that every old household should sell. Fertility is fixed in this comparison; children affect how much space is left for the young household's adults.

A simpler transfer. The planner can also pay a small premium for each unit sold. Choose a uniform per-unit seller premium a with $0 < a < MV^Y - qr_t - K_{ijt}$. Replace $L(\epsilon)$ by $a\epsilon$, financed by the buyer. The old seller now gains to first order by $a\epsilon/c_j^2$, while the buyer still gains. For several pairs, the same premium works under a uniform positive surplus bound and uniform slackness and curvature bounds. An ordinary market-price sale with zero premium leaves the old seller indifferent only to first order and generally worse off at second order. The choice between exact compensation and this simpler premium affects how much information the planner needs about each seller.

Fixed moving costs. A positive fixed cost $f\mathbf{1}\{\epsilon > 0\}$ violates $C(0+) = 0$. The marginal theorem does not cover it. A sufficient finite-trade test instead requires feasible positive consumption and:

$$qr_t\epsilon + L(\epsilon) + C(\epsilon) < qc_i^2\left[1 - \left(1 + \frac{\epsilon}{s_i}\right)^{-\alpha/\beta}\right]. \tag{17}$$

This is the condition for a positive utility gain in (25). The total benefit from the move must cover its fixed cost. The marginal comparison alone does not establish this.

6 A conditional fertility prediction

Let $\mathcal{R}$ be the lifetime resources left after housing and compensating payments, so that $(1+k)x+\chi n = \mathcal{R}$. The marginal net payment for additional housing is $p = -\mathrm{d}\mathcal{R}/\mathrm{d}h$. For this fertility comparison, the young household reoptimizes its other choices after receiving the additional housing and paying the bill. Prices and the seller's fertility remain fixed. Implicit differentiation gives:

$$\frac{\mathrm{d}n}{\mathrm{d}h} = \frac{\alpha\kappa/s^2 - \chi p/((1+k)x^2)}{\Delta}, \qquad \Delta = \frac{\vartheta}{n^2} + \frac{\chi^2}{(1+k)x^2} + \frac{\alpha\kappa^2}{s^2} > 0. \tag{18}$$

More space raises fertility if its benefit dominates the resources used to pay for the additional housing. The payment includes compensation and real costs when those are charged to the young household.

Proposition 2 (Fertility and the net payment)**.** *For the reduced problem above, fertility increases locally exactly when $(1+k)\kappa(MV^Y)^2 > \alpha\chi p$. If $\kappa u/\chi > \alpha/(1+k)$ and the young household is housing constrained, fertility increases for any marginal payment $p < MV^Y$ that leaves it better off. This includes a pure relaxation of the cap with $p = u$ and the compensated marginal allocation whenever $p = u + K_{ijt} < MV^Y$.*

The proof is in Appendix A.

The sufficient condition uses user cost but no borrowing multiplier. The next result removes that price in a stationary specialization. Positive saving, the stated old-age choices, and strict down-payment binding are still required.

6.1 A sufficient condition entirely in primitives

At stationary prices, set $d = 1 - q + q\tau^p > 0$, so $u = dP$. If the down-payment limit alone binds, $\widehat{h} = b/[(1-\phi)P]$. The lifetime share spent on housing at this limit is independent of P when $\widetilde{w}$ is exogenous:

$$a_h = \frac{u\widehat{h}}{\widetilde{w}} = \frac{db}{(1-\phi)\widetilde{w}}. \tag{19}$$

For interior old choices, zero property tax, and the corresponding zero common rebate, $\widetilde{w} = y + b$. This is an explicitly labeled specialization of the property-tax model, not an assumption that its endogenous rebate is fixed in general equilibrium.

Proposition 3 (A price-free sufficient restriction)**.** *Maintain the reduced problem, stationary prices, and a strictly binding down-payment limit; saving remains positive and old choices remain interior. If $\widetilde{w}$ is exogenous and*

$$\frac{db}{(1-\phi)\widetilde{w}} \geq \frac{\alpha}{1+k+\alpha}, \tag{20}$$

then $\kappa u/\chi > \alpha/(1+k)$. Consequently a marginal cap relaxation at fixed prices and resources raises fertility. More generally, any local increase in housing with net marginal payment $p < MV^Y$ raises fertility while the same constraints bind. In the zero-property-tax, interior-old specialization the restriction is

$$\frac{(1-q)b}{(1-\phi)(y+b)} \geq \frac{\alpha}{1+\beta(1+\gamma+\omega_B)+\alpha}. \tag{21}$$

It contains neither prices nor multipliers.

The proof is in Appendix A.

This condition excludes households with very little liquid wealth relative to lifetime resources. A household example satisfying all the original owner constraints is given in Appendix A. The appendix also gives a negative fertility response outside the condition and corrects the derivative when a rental limit changes both young and old housing. The price-free restriction concerns a stationary household comparison; prices, transfers, and tenure can also change in a policy equilibrium.

7 Population comparisons and units

For two paths starting with the same young cohort, iteration of the cohort law gives an exact transition comparison:

$$\frac{Y_t^1}{Y_t^0} = \prod_{s=0}^{t-1} \frac{\bar{n}_s^1}{\bar{n}_s^0}. \tag{22}$$

Thus cumulative relative fertility determines relative young-cohort size; the old-cohort comparison follows with a one-period lag. This identity does not prove that a policy raises fertility in every period or that either path converges.

At stationary prices and transfers, the household problems determine average fertility and housing for each candidate (P, T). A steady state solves the replacement condition $\bar{n}(P, T) = 1/\nu$ and the rebate condition $T = q\tau^p P(\bar{h}^Y(P, T) + \bar{h}^O(P, T))/2$. Existence of such a root is a separate requirement; the identities below do not establish it.

At a positive stationary population, fertility must be $\bar{n}^* = 1/\nu$ and $Y^* = O^*$. Define $\bar{h}^*_{\text{hh}} = (\bar{h}^{Y*} + \bar{h}^{O*})/2$, housing per adult household. Fixed-stock housing clearing gives:

$$N^*_{\text{hh}} = Y^* + O^* = \frac{\bar{H}}{\bar{h}^*_{\text{hh}}}. \tag{23}$$

Two policies can therefore have the same replacement fertility but different terminal population levels. A larger level requires less housing per household at the terminal allocation when supply is fixed.

Child units. Let $n^L = a_n n$ be literal children per household. The equivalent demographic coefficient is $\nu^L = \nu/a_n$, so replacement literal fertility is a_n/ν. The equivalent household requirements are $\chi^L = \chi/a_n$ and $\kappa^L = \kappa/a_n$; replacing $\log n$ by $\log n^L$ changes utility only by a choice-independent constant. Thus the rescaling preserves behavior when all relevant objects are transformed together. With $\nu = 2$, replacement is $n = 0.5$. A conversion $a_n = 2.1$ gives 1.05 literal children, not 2.1; to map 0.5 to 2.1 requires $a_n = 4.2$ and $\nu^L = 1/2.1$. This is a change of units, not a choice of demographic parameters.

Households and resident persons. N_{hh} counts adult household units. Converting it to resident persons requires the number of adults per household and the timing and residence of children. If these are fixed coefficients a_Y, a_O and children are counted as dependents in their birth period, a consistent illustration is $N_{\text{persons,t}} = a_Y Y_t + a_O O_t + a_n \bar{n}_t Y_t$. At a positive steady state the ratio $N^*_{\text{persons}}/N^*_{\text{hh}}$ is $(a_Y + a_O + a_n/\nu)/2$. Fixed common coefficients preserve the ordering of terminal household and person counts. Their empirical values and the within-period counting convention are not selected by this illustration.

A stationary housing-supply extension. If the terminal stock is $S(P^*) > 0$, the scale identity becomes $N^*_{\text{hh}} = S(P^*)/\bar{h}^*_{\text{hh}}$. Hence the exact comparison is:

$$\log \frac{N^*_{\text{hh},1}}{N^*_{\text{hh},0}} = \log \frac{S(P^*_1)}{S(P^*_0)} - \log \frac{\bar{h}^*_{\text{hh},1}}{\bar{h}^*_{\text{hh},0}}. \tag{24}$$

When the only fiscal transfer is the property-tax rebate, $T^* = q\tau^p P^* \bar{h}^*_{\text{hh}}$, the housing-stock scale cancels from the stationary household and fiscal equations. Supply may then change terminal population without changing their stationary price roots. This observation requires supplier profits, land income, and construction financing not to add new household-income or fiscal effects. A supply curve alone does not define the transition law or its resource costs.

8 Transitions after a housing policy

A higher borrowing limit lets some young households buy more housing. Prices and transfers then adjust, and changes in fertility alter the size of later cohorts. These responses must be solved together. The household result above compares the value of space with its cost. These additional equilibrium responses also affect fertility.

Removing the gains tax removes one source of kinks. The transition argument also requires smooth aggregate choices: this must be checked when households cross a housing or borrowing limit. Smoothness alone does not ensure that a path exists or converges. Near a steady state, the conditions in the appendix give a converging path from nearby initial states. A finite reform can be covered by following that path as the reform is increased, provided no solution reaches a feasibility boundary or loses its local uniqueness. The precise statement and short continuation proof are in the appendix. The conditions concern feasibility and local uniqueness along the whole policy change.

8.1 An illustrative credit reform

Consider an unexpected permanent increase in the financed share from 80%, starting from the same steady state in each comparison. Property tax remains 5%, preferences are fixed, and housing supply is fixed. The numbers below are a constructed example of the theory; they are not estimates for an economy.

Financed share	Initial fertility (model units)	Final price	Final households (initial = 100)
80%	0.5000	0.3427	100.00
81%	0.5019	0.3559	101.43
85%	0.5039	0.4152	105.97
90%	0.4992	0.4457	106.43

All three reforms lead to a larger terminal household population. Fertility returns to replacement, 0.5, in each steady state. It need not rise at every date: the 90% reform initially lowers average fertility slightly, and all three paths briefly fall below replacement later in the transition.

In the 90% case, fertility among owners rises initially from 0.5495 to 0.5764, but the owner share falls from 74.86% to 65.83%. Renters have lower fertility, which also falls slightly. Holding initial tenure shares fixed would raise average fertility by 0.01956 model units; the change in tenure shares subtracts 0.02040. Their sum explains the initial decline. This is an accounting decomposition, not a causal separation of the price and credit effects. At 90% financing the owner down-payment constraint is slack at the reported dates, so the strictly binding-cap derivative cannot describe that whole reform.

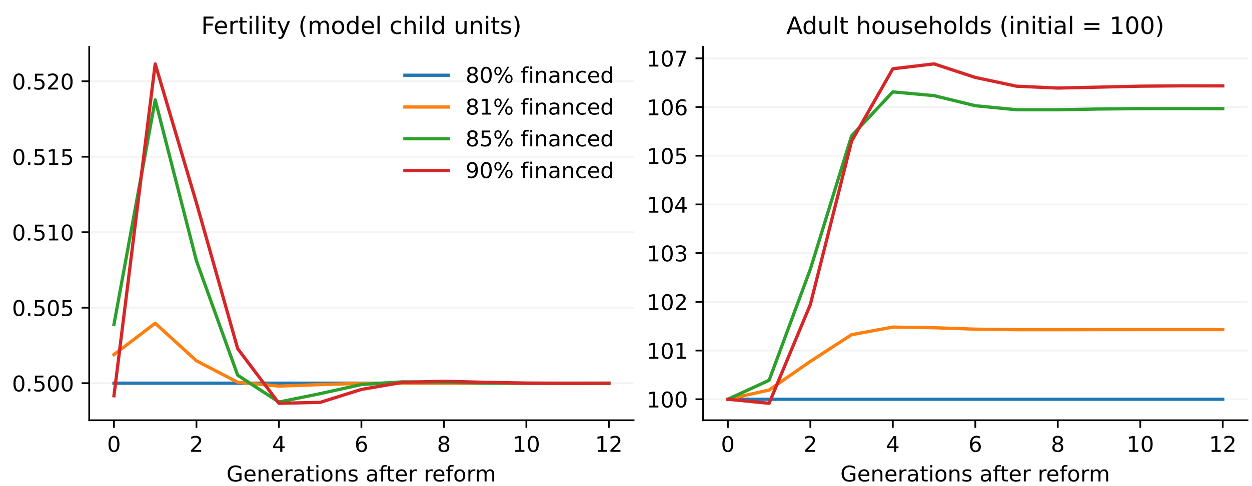

Figure 1: Fertility and population after a permanent credit reform. A period is a generation, not a year. All paths begin with the same young and old cohort sizes; date 0 is the reform date.

The example reallocates housing toward the young. Under 85% financing, terminal housing per young household rises from 0.7685 to 0.8066, while old housing falls from 0.6058 to 0.4903. Average housing per household falls, allowing the fixed stock to accommodate a larger population. The credit reform itself has not been shown to make every household better off; Proposition 1 concerns a compensated transaction.

The computed paths satisfy the household budgets, housing clearing, government budget, and cohort laws to small numerical errors. A longer horizon and a second solution method give the same 85% path to the reported precision; direct optimization of the original household problem also agrees with the reduced choices. Appendix C records the parameters and errors. These checks support the examples. They do not verify every condition of the finite-reform theorem throughout a policy interval.

Heterogeneity. The two-period structure also limits how much history must be carried forward. For cohorts generated by this model, entrant income and liquid wealth have a fixed distribution. Once the prices, transfers, and policy rules facing a cohort when young are known, its distribution of old assets and housing is determined. Thus heterogeneous entrants do not by themselves require an unrestricted distribution as a dynamic state here. This argument uses the model's fixed entrant-wealth distribution and two-period life; actual inheritance or longer lives can change that conclusion.

A Household derivations and the compensated sale

A.1 Proof of Proposition 1

The payments in the compensated sale are:

Party	Date t	Date $t+1$
Young buyer	Borrows $(P_t + q\tau^p P_t)\epsilon + L(\epsilon) + C(\epsilon)$; buys title, places $q\tau^p P_t\epsilon$ in a bond reserve, pays compensation and real cost.	Sells the additional unit for $P_{t+1}\epsilon$ and repays the loan; the reserve returns $\tau^p P_t\epsilon$ and pays the tax.
Old seller	Sells ϵ, receives $P_t\epsilon+L$; consumes $c_j^2(h_j^2/(h_j^2-\epsilon))^\gamma$; keeps the estate fixed using the bond and its lower property-tax obligation.	Estate spending is unchanged. The estate no longer sells this unit.
Lenders	Advance the loan at the market bond price.	Receive principal times $1/q$; no loss.
Government	Ownership changes; the taxable stock does not.	Total property-tax receipts and common rebates are unchanged.

The seller sets aside $qP_{t+1}\epsilon$ of the sale proceeds in a bond. This replaces the housing sold from its estate. It also saves $q\tau^p P_t\epsilon$ in the amount set aside for property tax. Together, the sale and tax saving leave $qr_t\epsilon$ for consumption. The buyer sells the additional unit when old, at the same date the seller's estate would have sold it. Housing use after date t can therefore stay unchanged. External borrowing finances the payments, and the loan is repaid in full.

Proof. The compensation in (16) leaves the seller's utility unchanged. Selling a small amount of housing relaxes its retention and estate constraints. For the buyer, keep young consumption, old housing, and the estate fixed, and pay the remaining bill from old consumption. The table shows that this reduces old consumption by $[qr_t\epsilon+L(\epsilon)+C(\epsilon)]/q$. For a sufficiently small sale, consumption stays positive and the other constraints remain slack. The buyer's lifetime utility gain is:

$$\Delta U_i = \alpha \log\left(1+\frac{\epsilon}{s_i}\right) + \beta \log\left(1-\frac{qr_t\epsilon + L(\epsilon) + C(\epsilon)}{qc_i^2}\right). \tag{25}$$

The interior saving condition gives $c_i^2 = \beta x_i/q$, so the right derivative at zero is $(MV^Y - qr_t - K_{ijt})/x_i > 0$. The buyer therefore gains from a sufficiently small sale. Property-tax receipts, future housing supply, and other households' allocations stay unchanged, and lenders receive the market return. □

A.2 Proof of Proposition 2

Proof. Differentiate $(1+k)x + \chi n = \mathcal{R}(h)$ to obtain $(1+k)x' = -p - \chi n'$. Differentiating $\vartheta/n = \chi/x + \alpha\kappa/(h-\kappa n)$ and substituting for x' gives (18). Multiply its numerator by the positive number $(1+k)x^2$ and use $MV^Y = \alpha x/s$. For the sufficient condition, $MV^Y \geq u$ implies $(1+k)\kappa MV^Y > \alpha\chi$, hence $(1+k)\kappa(MV^Y)^2 > \alpha\chi p$ whenever $p < MV^Y$. □

A.3 Proof of Proposition 3

Proof. Let $D = 1 + k + \alpha + \vartheta$ and $v = \kappa u/\chi$. Without the current cap, the conditional problem spends the following fraction of resources on housing:

$$g(v) = \frac{\alpha + \vartheta v/(1+v)}{D}. \tag{26}$$

Strict concavity implies that the current cap binds strictly exactly when $a_h < g(v)$ in this reduced problem. The function g is strictly increasing, and direct substitution gives $g(\alpha/(1+k)) = \alpha/(1+k+\alpha)$. Therefore (20) and strict binding imply $g(v) > g(\alpha/(1+k))$ and hence $v > \alpha/(1+k)$. Proposition 2 supplies the fertility implication. □

The condition can hold while the cap binds strictly: the housing expenditure share can lie in $\alpha/(1+k+\alpha) \leq a_h < (\alpha+\vartheta)/D$. This interval has positive length. The condition still excludes households with very little liquid wealth relative to lifetime resources. A constructed example also satisfies the original owner constraints. Set $q = 0.8$, $\phi = 0.7$, $P = 4$, $y = 4.8$, $b = 2.4$, and $\tau^p = T = 0$. Let $\beta = 0.8$, $\gamma = 0.5$, $\omega_B = 3$, $\alpha = \chi = \kappa = 1$, and $\vartheta = 2$. Then $1 + k = 4.6$, $u = 0.8$, $h = 2$, and $x = s = n = 1$. The share $a_h = 2/9$ exceeds $\alpha/(1+k+\alpha) = 5/28$, while $MV^Y = 1 > u$. Saving is 2.8, old consumption is 1, old housing is 0.625, and the estate is $3.75 > 4(0.625)$. Thus positive saving and the two old-age slackness conditions are consistent with the restriction. The cap derivative is $19/74 > 0$. This is a conditional household example, not a complete equilibrium construction.

A.4 A negative response and a common rental limit

More housing can improve welfare and lower fertility. Set $1+k = 2$, $\alpha = \chi = x = s = n = 1$, $\kappa = 0.1$, $\vartheta = 1.1$, and $u = 0.5$. Then $h = 1.1$, $\widetilde{w} = 3.55$, $MV^Y = 1$, and $\Delta = 1.61$. At $p = u$, the household is better off with more housing but $\mathrm{d}n/\mathrm{d}h = -15/161$. If the added space is provided at zero marginal net payment, the same conditional derivative is $10/161 > 0$. This is a constructed household example, not a calibration or an equilibrium counterfactual.

A common rental limit changes future resources. If both the young and old rental caps bind, raising their common limit changes current space and the old household's committed rent. At fixed prices and transfers its total derivative is:

$$\frac{\mathrm{d}n}{\mathrm{d}h_R^{\max}} = \left.\frac{\partial n}{\partial \widehat{h}}\right|_{\widetilde{w}} - q^2 r_{t+1}\frac{\partial n}{\partial \widetilde{w}}, \qquad \frac{\partial n}{\partial \widetilde{w}} = \frac{\chi}{(1+k)x^2\Delta} > 0. \tag{27}$$

Reporting the first term alone overstates the fertility response to the common cap change under these constraints.

B Conditions for a converging transition

For each reform, we need initial prices that generate a feasible path to the new steady state. The conditions below allow us to follow those prices as the reform grows. We follow one continuous family of steady states; other steady states may exist.

Let $\lambda \in [0,1]$ index the size of a permanent reform, with $\lambda = 0$ the initial policy. Let $s_t \in \mathbb{R}^d$ describe inherited cohort sizes and household choices, and let $j_t \in \mathbb{R}^k$ contain prices not fixed by those inherited choices. Write $z_t = (s_t, j_t)$. Suppose the equilibrium equations locally determine next period's variables as a twice continuously differentiable map, jointly smooth in the policy and the state:

$$z_{t+1} = g_\lambda(z_t), \qquad g_\lambda(z^*(\lambda)) = z^*(\lambda). \tag{28}$$

Aggregate differentiability requires either uniformly regular household choices with unchanged constraints or a separate argument permitting differentiation of the integrals across changing household choices. This representation requires a nonsingular derivative of the one-date equations with respect to next-date variables after redundant and static equations have been removed. It is an assumption to check, not a consequence of the two-period structure alone.

The local condition. Suppose the derivative of g_λ at the steady state has d eigenvalues strictly inside the unit circle and k strictly outside it. Suppose also that the stable subspace projects with full rank onto the inherited coordinates. The local stable-manifold theorem then gives a function $j = \Gamma_\lambda(s)$: initial points on this graph remain close to the steady state and converge to it. Choose the local graph inside a feasible neighborhood and forward invariant there, so every point on it supplies a feasible converging tail. The conclusion permits zero stable eigenvalues; only the unstable part is inverted in the proof. The version for possibly noninvertible maps is covered by McGehee and Sander (1996), Theorem 1. Joint smoothness in the policy and the stated strict conditions give a continuously differentiable graph locally in λ.

Proposition 4 (Continuation to a finite reform)**.** *Hold $s_0 = \Pi_s z^*(0)$ fixed and let $j_0^* = \Pi_j z^*(0)$. Suppose a continuous family of positive steady states $z^*(\lambda)$ has the local stable graphs just described. There are a finite date J and a bounded open set $B \subset \mathbb{R}^k$ of initial prices with the following properties. For every $(\lambda, j) \in [0,1] \times \overline{B}$, the first J iterations from (s_0, j) are defined and feasible, and their inherited coordinates lie in the domain of the final stable graph. Define their deviation from this graph by:*

$$\mathcal{F}(\lambda, j) = \Pi_j g_\lambda^J(s_0, j) - \Gamma_\lambda\Big(\Pi_s g_\lambda^J(s_0, j)\Big)\,. \tag{29}$$

Here Π_s and Π_j select inherited variables and prices. Assume $\mathcal{F}$ is continuously differentiable on an open neighborhood of $[0,1] \times \overline{B}$, has no zero on $[0,1] \times \partial B$, and its derivative with respect to j is nonsingular at every zero. At $\lambda = 0$, assume $j_0^* \in B$ is its only zero. Then for every $\lambda \in [0,1]$ there is exactly one such initial price vector in B. It varies continuously differentiably with λ and generates an infinite equilibrium path converging to $z^*(\lambda)$.

Proof. At any zero, the implicit-function theorem continues the initial prices uniquely for nearby policy values. The zeros form a compact set because $\mathcal{F}$ is continuous, $\overline{B}$ is compact, and zeros cannot lie on its boundary. For a fixed policy, nonsingularity makes each zero isolated; compactness then makes their number finite. Near any policy value, each zero continues and no new zero can appear elsewhere: otherwise a sequence of such zeros would have a limit at that policy, contradicting its local uniqueness. The number of zeros is therefore locally constant on $[0,1]$. It equals one at zero, so it equals one throughout the interval. Patching the local functions gives the claimed smooth initial prices. At date J the corresponding orbit lies on the local stable graph, which supplies the converging infinite tail. □

The reform need not be small. The restrictions instead prevent a path from reaching a feasibility boundary, prevent two solutions from merging, and require the tail to enter the region where local convergence is established. They have not been reduced to primitive inequalities for the full model. The result does not imply that all equilibria outside B are absent. Checking only a few terminal horizons or the eigenvalues at one endpoint is insufficient to verify these assumptions for a whole policy interval.

Why a finite description of the old cohort is possible. Let $i = (y, b)$ be a young household's type, drawn from the fixed distribution F. Let η_t collect the finite list of prices, transfers, and policy parameters used in its lifetime problem at date t, including $(P_t, P_{t+1}, P_{t+2}, T_t, T_{t+1})$ and the current and anticipated policy rules. Assume its optimal choices are unique, measurable in type, and integrable. Its optimal financial assets and owner housing are functions $a_R(i;\eta_t)$, $a_O(i;\eta_t)$, and $H_O(i;\eta_t)$, and its ownership probability is $\pi^O(i;\eta_t)$. For any bounded test function f, the next old-cohort distribution is given by:

$$\int f \,\mathrm{d}G_{t+1} = \int \big[(1-\pi^O(i;\eta_t)) f(a_R(i;\eta_t), 0, R) \\ + \pi^O(i;\eta_t) f(a_O(i;\eta_t), H_O(i;\eta_t), O)\big] \,\mathrm{d}F(i). \tag{30}$$

Thus the model-generated old distribution is indexed by η_t, even when F is continuous. This describes distributions generated by the model, rather than arbitrary initial distributions. At an unexpected reform, the inherited distribution uses the expectations under which the old cohort actually chose when young: $G_0 = K(\eta_{-1}^{\mathrm{pre}})$, where K is the distribution defined by (30). Later cohorts use the prices and policy rules anticipated when they make their young-period choices. This finite representation does not guarantee smooth aggregate choices or a nonsingular next-date derivative; those remain separate conditions in the transition argument.

C Parameters and numerical checks

The example has identical entrant income and liquid wealth, with heterogeneity only in the ownership taste. All parameters are chosen for illustration:

Object	Value
Bond price and property tax	$q = 0.5$, $\tau^p = 0.05$
Housing, space, and goods parameters	$\alpha = 0.4$, $\kappa = 0.5$, $\chi = 0.15$
Lifetime and old-age weights	$\beta = 0.4$, $\gamma = 0.3$, $\omega_B = 0.4$
Fertility weight	$\vartheta = 0.35$ throughout
Entrant income and liquid wealth	$y = 1$, $b = 0.06$
Demography and fixed housing stock	$\nu = 2$, $\bar{H} = 2$
Rental and owner size limits	$h_R^{\max} = 0.45$, $h_O^{\max} = 2$
Ownership taste	$\bar{\xi} = -0.15$, $\sigma_\xi = 0.12$
Financed share	Initial 0.80; final 0.80, 0.81, 0.85, 0.90

Prices and transfers are set to their final steady-state values after date 28. Household states at that date are generated by the preceding choices, not imposed at the endpoint. Across the four paths, the maximum housing error is 3.45×10^{-10} units and the maximum government-budget error is 3.85×10^{-12} goods. The largest terminal-state discrepancy, scaled by the larger of one and the relevant steady-state value, is 2.93×10^{-10}.

For the 85% policy, extending the terminal date from 28 to 40 changes reported prices, transfers, fertility, and cohort sizes by at most 2.13×10^{-13} over their common dates. A second least-squares solution with a separately written market and fiscal residual agrees with the original log price and transfer paths to 1.10×10^{-12}.

The original two-budget household problem was also optimized directly for both tenures at dates 0, 2, and 28 under every policy: 24 comparisons. The largest difference in the seven choice variables is 1.07×10^{-6}. This checks the saving and old-age constraints independently of the reduced fertility equation. Original-equation residuals and binding constraints are checked at every computed date; the independent optimization comparison covers dates 0, 2, and 28.

At each stationary endpoint, including the initial policy, the numerical linearization has seven stable and two unstable roots after eliminating the static transfer equation. Its stable directions project with full rank onto the inherited variables. The redundant acquisition-basis state is harmless when the gains tax is zero and contributes a zero root. These are local numerical diagnostics, not rigorous bounds over a policy interval. In particular, the owner borrowing constraint changes from binding to slack between the smaller reforms and the 90% case.

The household sign checks cover 4,280 binding-cap configurations, including 796 satisfying the primitive sufficient condition. Every qualifying case has a positive response; 548 other cases have a negative response. Analytical and finite-difference derivatives differ by at most 6.69×10^{-8}. The financing checks cover 108 transactions, and the child rescaling, cohort-product formula, tenure decomposition, and stationary supply identities are checked separately. These calculations supplement the proofs; numerical agreement is not used to replace a mathematical assumption.